MSG No. 20.36 C_a222_1 [Type IV, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `2..`

No.	position	mapping
1	$[x,\, 0,\, 0]$	`[1,2]`
2	$[-x,\, 0,\, \frac{1}{2}]$	`[3,4]`
3	$[x+\frac{1}{2},\, \frac{1}{2},\, 0]$	`[5,6]`
4	$[\frac{1}{2}-x,\, \frac{1}{2},\, \frac{1}{2}]$	`[7,8]`
5	$[x+\frac{1}{2},\, 0,\, 0]$	`[9,10]`
6	$[\frac{1}{2}-x,\, 0,\, \frac{1}{2}]$	`[11,12]`
7	$[x,\, \frac{1}{2},\, 0]$	`[13,14]`
8	$[-x,\, \frac{1}{2},\, \frac{1}{2}]$	`[15,16]`

Table 2: Wyckoff site: `8b`, site symmetry: `2'..`

No.	position	mapping
1	$[x,\, \frac{1}{4},\, 0]$	`[1,14]`
2	$[x,\, \frac{3}{4},\, 0]$	`[2,13]`
3	$[-x,\, \frac{1}{4},\, \frac{1}{2}]$	`[3,16]`
4	$[-x,\, \frac{3}{4},\, \frac{1}{2}]$	`[4,15]`
5	$[x+\frac{1}{2},\, \frac{3}{4},\, 0]$	`[5,10]`
6	$[x+\frac{1}{2},\, \frac{1}{4},\, 0]$	`[6,9]`
7	$[\frac{1}{2}-x,\, \frac{3}{4},\, \frac{1}{2}]$	`[7,12]`
8	$[\frac{1}{2}-x,\, \frac{1}{4},\, \frac{1}{2}]$	`[8,11]`

Table 3: Wyckoff site: `8c`, site symmetry: `.2.`

No.	position	mapping
1	$[0,\, y,\, \frac{1}{4}]$	`[1,3]`
2	$[0,\, -y,\, \frac{3}{4}]$	`[2,4]`
3	$[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{4}]$	`[5,7]`
4	$[\frac{1}{2},\, \frac{1}{2}-y,\, \frac{3}{4}]$	`[6,8]`
5	$[\frac{1}{2},\, y,\, \frac{1}{4}]$	`[9,11]`
6	$[\frac{1}{2},\, -y,\, \frac{3}{4}]$	`[10,12]`
7	$[0,\, y+\frac{1}{2},\, \frac{1}{4}]$	`[13,15]`
8	$[0,\, \frac{1}{2}-y,\, \frac{3}{4}]$	`[14,16]`

Table 4: Wyckoff site: `8d`, site symmetry: `.2'.`

No.	position	mapping
1	$[\frac{1}{4},\, y,\, \frac{1}{4}]$	`[1,11]`
2	$[\frac{1}{4},\, -y,\, \frac{3}{4}]$	`[2,12]`
3	$[\frac{3}{4},\, y,\, \frac{1}{4}]$	`[3,9]`
4	$[\frac{3}{4},\, -y,\, \frac{3}{4}]$	`[4,10]`
5	$[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{4}]$	`[5,15]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[6,16]`
7	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[7,13]`
8	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[8,14]`

Table 5: Wyckoff site: `16e`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x,\ -y,\ -z]$	`[2]`
3	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[3]`
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[4]`
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[5]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[6]`
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[7]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[8]`
9	$\left[x+\frac{1}{2},\ y,\ z\right]$	`[9]`
10	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	`[10]`
11	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	`[12]`
13	$\left[x,\ y+\frac{1}{2},\ z\right]$	`[13]`
14	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	`[14]`
15	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[15]`
16	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[16]`